$X_1, X_2, \ldots, X_5$ iid Unif. on $\{0, 1, \ldots, 15\}$

$$P\left(\sum_{i=1}^{5} X_i = 15\right)$$

① Using CLT with "continuity correction"

$$P\left(\sum_{i=1}^{5} X_i = 15\right) \approx P(14.5 \leq Y \leq 15.5) = 0.0035788$$

where $Y \sim N\left((7.5)(5), (5)\underset{106.25}{V(X_i)}\right)$

② 16^5 possible values for $X_1, X_2, \ldots, X_5$, they are ELO

How many of these have $\sum X_i = 15$?

First note that

$X_1 = 15, \; X_2 = X_3 = X_4 = X_5 = 0$

$X_1 = X_2 = X_3 = X_4 = 0, \; X_5 = 15$

are two different possibilities

Hence, the numerator is <u>not</u> $\binom{15}{5}$

Instead, imagine 19 spaces :

Place a "divider" | in 4 of them

Place a "O" in the other 15

The result will be 5 groups of circles, their total has to be 15

X_1 = # of "O" to the left of first divider

X_2 =

There are $\binom{19}{4}$ ways of placing the "dividers"

So, prob. is $\binom{19}{4} / 16^5$

$X \sim N(0, \sigma^2)$

$E(|X|) = \sigma\sqrt{\frac{2}{\pi}}$

Efficiency of the MLE

It is possible to show that the lower bound on the variance of any unbiased estimator is $1/nI(\theta_0)$. This is called the Cramer-Rao lower bound.

Since the MLE is asymptotically unbiased, we know that we are achieving the Cramer-Rao lower bound, at least asymptotically.

Hence, we are minimizing the MSE (asymptotically). This is objectively the best rationale for utilizing maximum likelihood estimators.

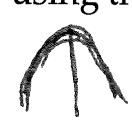
Numerical Approach

In cases where the likelihood is maximimized numerically, it is unlikely that the Fisher Information can be derived.

Instead, we will use an approximation based on the result seen above:

$$nI(\theta) = E\left[-\frac{\partial^2}{\partial\theta^2}\log L(\theta)\right]$$

This result suggests that we can approximate $nI(\theta_0)$ by using the **curvature at the peak of the observed likelihood function**.



This quantity is returned by most numerical optimization routines, called the Hessian.

We will call this the observed Fisher Information.

Example: To illustrate the idea, let's consider a one-parameter model based on the Gamma distribution. In particular, we assume that $X_1, X_2, \ldots, X_n$ are i.i.d. Gamma$(\alpha, 1)$.

β

This means that

$$f_X(x;\alpha) = \frac{1}{\Gamma(\alpha)} x^{\alpha-1} e^{-x}, \;\; x > 0$$

If x is a vector holding the observed data values, then

```
> sum(dgamma(x,alpha,1,log=T))
```

β

would return the likelihood evaluated at alpha.

log

The following function will return the MLE and the Hessian.

```
> onedgamma = function(x)
+ {
+ # The negative of the log-likelihood.
+    negloglikelihood = function(theta,x)
+    {
+       return((-1)*sum(dgamma(x,theta,1,log=T)))
+    }
+
+ # Starting value is the MOM estimate.
+    thetahatmom = mean(x)
+
+ # The optimization.
+    optimout = optim(thetahatmom, negloglikelihood,
+                     x=x, hessian=T, method="BFGS")
+
+    return(list(mle = optimout$par, hessian=optimout$hessian))
+ }
```

Now, let's try it out:

```
> set.seed(1)
> x = rgamma(100, 3, 1)
> onedgamma(x)
$mle
[1] 3.022577
```

$\hat{\alpha}$

```
$hessian
         [,1]
[1,] 39.14847
```

$nI(\hat{\alpha})$

Conclude that our best estimate of α is 3.02, and we can say that the estimator is approximately normal with an approximate variance of $1/39.14 = 0.025$. The standard error is approximately 0.16.

$1/nI(\hat{\alpha})$

More interesting examples will be considered when we move to multi-dimensional parameters.

Exercise: Why did I not need to take the negative of the Hessian in order to find the variance?

Remember that we are minimizing the negative log likelihood. Can think: Already include (-1) when calculated the negative log likelihood

Instead of maximizing

we are minimizing

Confidence Intervals

In those cases where we can state that the MLE is approximately normal with mean θ_0 and variance $1/nI(\theta_0)$, we can report an approximate $100(1-\alpha)\%$ confidence interval for θ as

$$\widehat{\theta} \pm z_{\alpha/2}\sqrt{1\Big/ nI(\widehat{\theta})}$$

$\hat{\Theta} \pm z_{\alpha/2} SE(\hat{\Theta})$

The derivation of this is much like what we saw in the previous Part, and derives from the fact that

$$\frac{\widehat{\theta}-\theta_0}{\sqrt{1\Big/ nI(\widehat{\theta})}}$$

$P(-z_{\alpha/2} \leq \quad \leq z_{\alpha/2}) = 1-\alpha$

has approximately the standard normal distribution.

$P(L \leq \theta_0 \leq U)$

Example: Suppose that $X_1, X_2, \ldots, X_n$ are i.i.d. Poison(λ). Based on our previous results, we can now derive an approximate $100(1-\alpha)\%$ confidence interval for λ as

$$\overline{X} \pm z_{\alpha/2}\sqrt{\overline{X}/n}$$

since the MLE for λ is $\overline{X}$ and $I(\widehat{\lambda}) = 1/\overline{X}$.

The validity of the above interval will depend on n. Is n large enough? The best way to check is via simulations, with realistic choices of λ_0 and n.

1 Consider the sample code below.

```
> # The true value of the parameter
> theta0 = 10
>    λ
> # The sample size
> n = 20
>
> # The number of repetitions of the simulation experiment
> reps = 10000
>
> # What confidence is desired? (i.e., we will construct
> # 100(1-alpha)% confidence intervals
> alpha = 0.05
>
> # Store thetahat, along with the lower and upper endpoint
> # of each confidence interval
> thetahat = numeric(reps)
```



```
> lower = numeric(reps)
> upper = numeric(reps)
>
> # Also store whether or not the confidence interval covers
> # the true value of the parameter
> coverstruth = logical(reps)
>
> # Start the loops
> for(i in 1:reps)
+ {
+
+ # Simulate the data set
+    x = rpois(n, theta0)
+
+ # Calculate the MLE
+    thetahat[i] = mean(x)
+
```

```
+ # Construct the confidence interval z_{α/2}
+    lower[i] = thetahat[i] - qnorm(1-alpha/2)*sqrt(thetahat[i]/n)
+    upper[i] = thetahat[i] + qnorm(1-alpha/2)*sqrt(thetahat[i]/n)
+
+ # Test if it covers theta0 or not
+    coverstruth[i] = (lower[i] < theta0) & (upper[i] > theta0)
                                            truth                 truth
+ }                 (L, U)
>
> # Print the proportion that covers the truth. This should be
> # close to 1-alpha
> print(sum(coverstruth)/reps)
[1] 0.9498
```


Exercise: For $\lambda_0 = 10$, for how small of values of n does the normal approximation still work adequately? What about when $\lambda_0 = 1$?

When $\lambda_0 = 10$, but $n=5$, still performs well, i.e. coverage is close to 0.95

When $\lambda_0 = 1$, but $n=5$, significantly undercover

Remember that the performance comes down to whether or $\bar{X}$ is approx. normal

CLT says that the sample size (n) required for $\bar{X}$ to be approx. normal is smaller when drawing from an approx. normal pop, such as Poisson(10)

The Delta Method

Often, our objective is not to estimate θ, but instead some function of θ, call it $g(\theta)$. For example, often want to estimate the probability of some event A. This probability is not a parameter, but it is a function of θ.

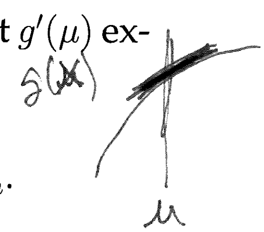
From the invariance property we know that the MLE of $g(\theta)$ is $g(\hat{\theta})$. But, we also need to calculate a SE or construct a confidence interval for this estimate.

One approach would be to "reparameterize" the model in terms of $\overset{\lambda =}{g(\theta)}$, and then find $I(\overset{\lambda}{g(\theta)})$, and then use $1/nI(\overset{\lambda}{g(\theta)})$ as the variance, etc. Fortunately, there is an easier way...

Recall the Delta Method:

In prob: $Y_n = \bar{X}_n$
$\sigma_n^2 = \sigma^2/n$

> If Y_n is approximately $N(\mu, \sigma_n^2)$ and $\sigma_n^2 \to 0$ as n increases, then $g(Y_n)$ is approximately $N(g(\mu), (g'(\mu))^2\sigma_n^2)$, assuming that $g'(\mu)$ exists and is not zero.



Here we will use this result with the MLE for θ in place of Y_n.

Then, $\mu = \theta_0$ and $\sigma_n^2 = 1/nI(\theta_0)$.

We can conclude that $g(\hat{\theta})$ is approximately

$$N\left(g(\theta_0),\ g'(\theta_0)^2\left(\frac{1}{nI(\theta_0)}\right)\right)$$

In practice, we use the approximate variance:

$$g'(\hat{\theta})^2\left(\frac{1}{nI(\hat{\theta})}\right)$$

Exercise: Suppose that $X_1, X_2, \ldots, X_n$ are i.i.d. Exponential(λ). We seek to estimate $\tau = P(X_i > a)$ for some constant $a > 0$. Construct the MLE and find its standard error. Also construct a confidence interval for τ. $\tau = g(\lambda) = e^{-\lambda a}$

By invariance property, the MLE for τ is

$$\hat{\tau} = g(\hat{\lambda}) = e^{-\hat{\lambda} a} = e^{-a/\bar{X}} \quad \text{since } \hat{\lambda} = 1/\bar{X}$$

We know $I(\lambda) = 1/\lambda^2$

$$g'(\lambda) = \frac{\partial g}{\partial \lambda} = -a e^{-\lambda a}$$

So, we can say that $\hat{\tau}$ is approx.

$$N\left(\tau_0, \left(g'(\lambda_0)\right)^2 \left(\frac{1}{n I(\lambda_0)}\right)\right)$$

i.e. $N\left(\tau_0, \left(a^2 e^{-2\lambda_0 a}\right)\left(\frac{\lambda_0^2}{n}\right)\right)$

Practical Result: $\hat{\tau}$ is approx. normal with variance of

$$\left(a^2 e^{-2a/\bar{X}}\right)\left(\frac{1}{n\bar{X}^2}\right)$$

and approx. SE of

$$\left(a e^{-a/\bar{X}}\right)\left(\frac{1}{\sqrt{n}\,\bar{X}}\right)$$

and a $100(1-\alpha)\%$ CI for τ is

$$\hat{\tau} \pm z_{\alpha/2}\left(a e^{-a/\bar{X}}\right)\left(\frac{1}{\sqrt{n}\,\bar{X}}\right)$$

When does this fall apart?

$X_1, X_2, \ldots, X_n$ iid $\text{Unif}(0,\theta)$

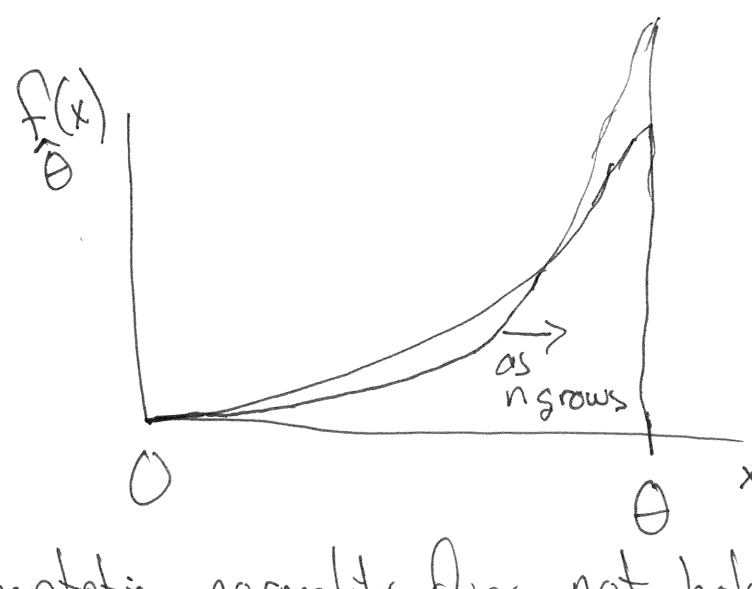
$\hat{\theta}_{MLE} = X_{(n)}$

What is the dist. of $\hat{\theta}_{MLE}$?

$$P(\hat{\theta}_{MLE} \leq x) = P(X_{(n)} \leq x) = \begin{cases} \left(\frac{x}{\theta}\right)^n, & 0 \leq x \leq \theta \\ 0, & x < 0 \\ 1, & x > \theta \end{cases}$$

So, the density is

$$f_{\hat{\theta}}(x) = \begin{cases} n\left(\frac{x}{\theta}\right)^{n-1}\left(\frac{1}{\theta}\right), & 0 \leq x \leq \theta \\ 0, & \text{otherwise} \end{cases}$$



never looks "normal" for any n

Asymptotic normality does not hold. The problem is that the support depends on θ, violating a "regularity conditions"